High Returns, Low Investment: Public Capital and the Euler Equation Puzzle

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Abstract

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1 Introduction

I will show some examples of things you can easily do in quarto-format.Results

2 An Euler Equation Model with Public Capital Externalities

I build on the intertemporal optimisation framework of Kremer et al. (2019). I first consider the case where capital is entirely public and investment occurs only through taxation, showing that this generates a collective action problem that dilutes the effective return to capital. I then extend the standard model to incorporate both public and private capital, allowing households to allocate investments between the two.

2.1 Baseline: Standard Euler Equation

Consider in a deterministic discrete-time intertemporal consumption model, households maximise their discounted lifetime utility over an infinite horizon.

Optimisation Problem

Households face the following optimisation problem:

$$\max_{\{C_t\}} \sum_{t=0}^{\infty} \delta^t u(C_t) \tag{1}$$

where $0 < \delta < 1$ is the annual discount factor, C_t is the household consumption in period t, and $u(C_t)$ measures the utility from consuming C_t in period t. $u(C)$ is increasing and concave and follows Inada Conditions: $u'(C) > 0$, $u''(C) < 0$, $\lim_{C\to 0} u'(C) = \infty$, and $\lim_{C\to\infty} u'(C) = 0$.

Budget Constraint

$$x_{t+1} = f'(K_t)(x_t - C_t) \tag{2}$$

x_t is household assets ('cash on hand') in period t with x_0 given, C_t is consumption in period t, and $f'(K_t)$ is the gross return to capital. $f(K)$ is increasing and concave, following: $f'(K) > 0$, $f''(K) < 0$, $\lim_{K\to 0} f'(K) = \infty$, and $\lim_{K\to\infty} f'(K) = 1$ (long-run steady state).

Discrete Time Euler Equation

Solving for optimal intertemporal consumption path:

$$u'(C_t) = \delta f'(K_t) u'(C_{t+1}) \tag{3}$$

Assume constant-elasticity-of-substitution (CES) utility:

$$u(C) = \begin{cases} \frac{C^{1-\sigma}}{1-\sigma} & \text{if } \sigma \neq 1 \\ \ln(C) & \text{if } \sigma = 1 \end{cases}$$

This means that the elasticity of intertemporal substitution is constant and independent of C_t. Therefore, the consumption growth rate is:

$$\frac{C_{t+1}}{C_t} - 1 = (\delta f'(K_t))^{\frac{1}{\sigma}} - 1 \tag{4}$$

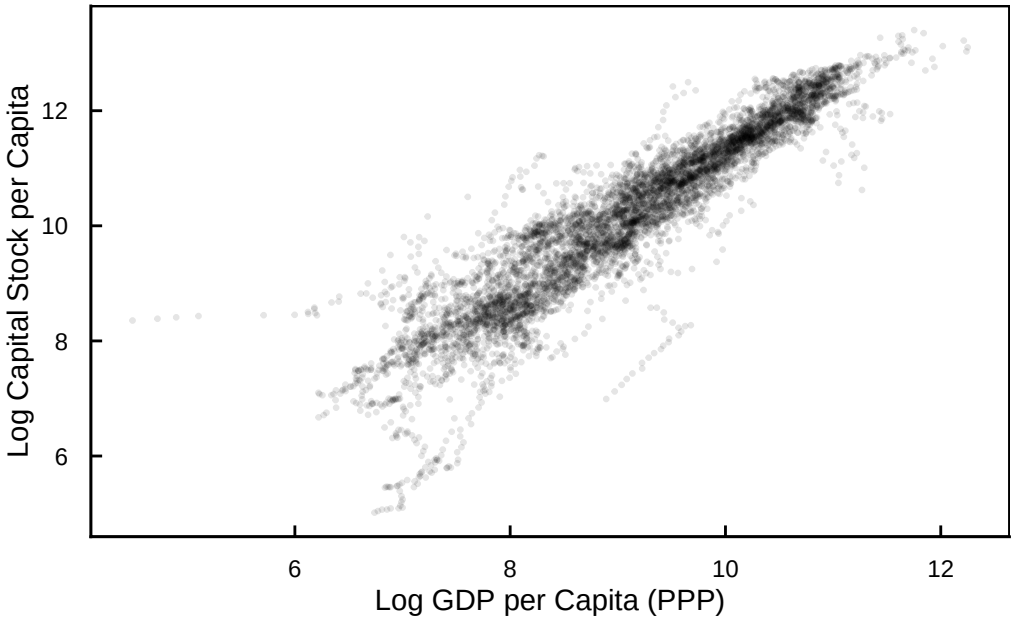
The Puzzle

To illustrate the puzzle, I use $\delta = 0.96$ [1]and $\sigma = 2$ [2] proposed in Kremer et al. (2019), and lagged internal rates of return from the Penn World Table (PWT) [3] (Feenstra et al., 2015) as a proxies for $f'(K)$ to model the predicted consumption growth rates under Equation 4. Intuitively, a concave production function means that countries with lower capital stock face higher returns on capital investment. Figure 1 shows that countries with lower income have lower capital stock. Therefore, from Equation 4, consumption growth rates should be high. However, Figure 2 Panel (A) shows that in many middle- and low-income countries, the actual growth rates fall below what the Euler Equation predicts. Moreover, Figure 2 Panel (B) shows that countries with higher predicted consumption growth rates tend to have larger gap between predicted and actual consumption growth rates. This pattern is most pronounced in countries such as Egypt, Venezuela, Côte d'Ivoire, and Zimbabwe. The puzzle, therefore, is why households in these economies invest at levels far below what the standard model predicts as optimal.

2.2 Capital as a Public Good

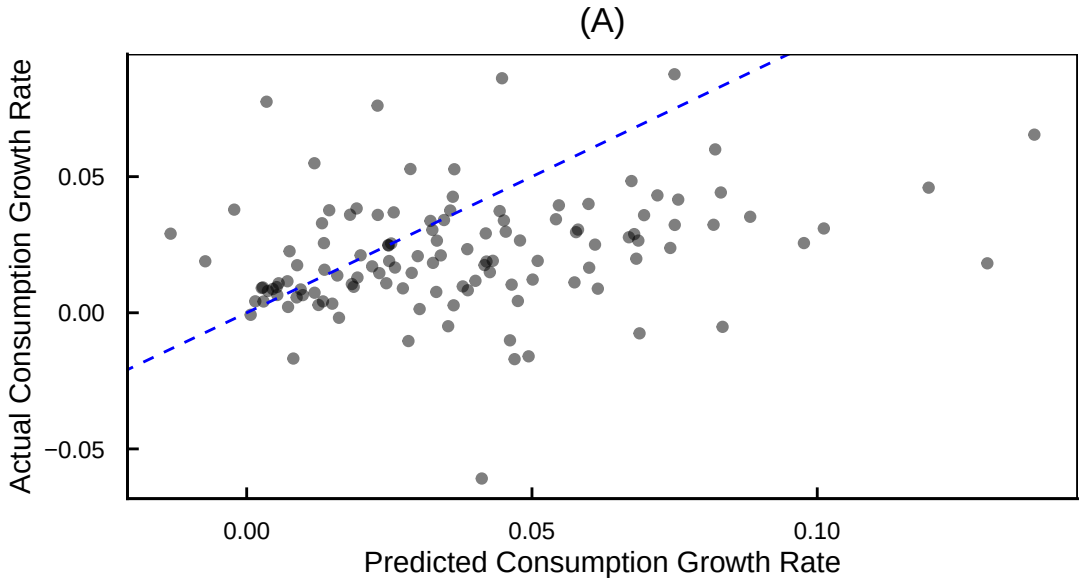
[1] $\frac{1}{\delta} - 1 \approx 0.04$, implying a 4% annual rate of time preference.
[2] Elasticity of intertemporal substitution of 0.5 is used in macro studies standard calibrations.
[3] Available for download at www.ggdc.net/pwt

Figure 1: Capital Stock and Income Across Countries 1950-2023

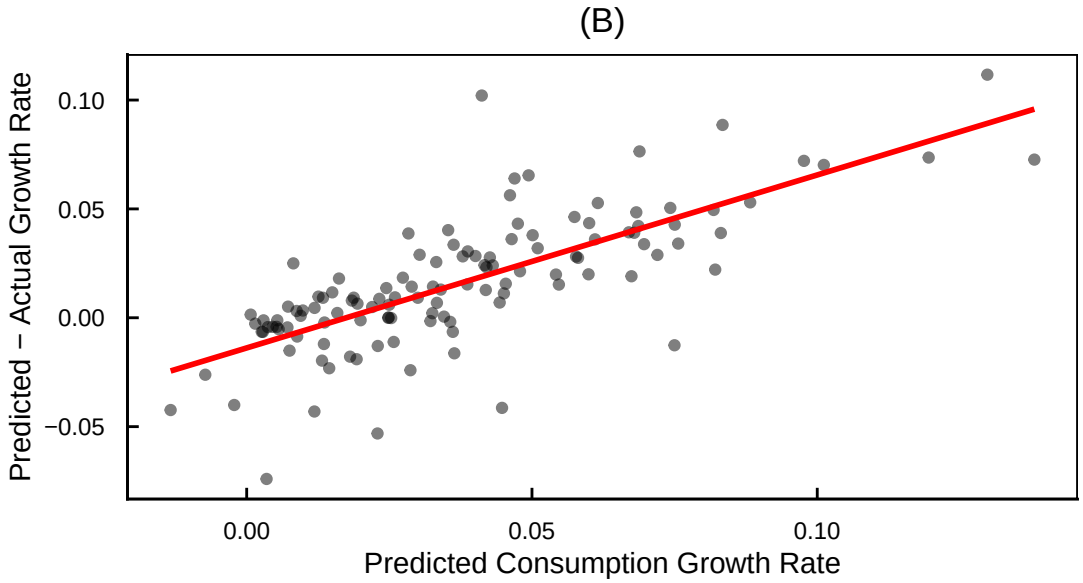


Notes: Capital stock at current PPPs.Expenditure−side real GDP at chained PPPs. Source: PWT 11.0.

Figure 2: Actual vs. Predicted Average Consumption Growth Across Countries 2010-2019



Notes: The dashed line is the 45 degree line. Countries with populations below 1 million and irregular internal rates of return excluded. Data after 2019 is excluded to avoid Covid−19 distortions. Source: PWT 11.0.



Notes: The red line is OLS fit. Source: PWT 11.0.

References

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